MILITARY SPECIFICATION

TRUCK, TANK; AIRCRAFT REFUELING, 5,000 GALLON, AVGAS OR JET FUEL, 6 BY 4, DED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a commercial, diesel-engine-driven, six wheel, four-wheel drive, 5,000-gallon capacity, aircraft fuel servicing tank truck.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-S-300	-	Sheeting and Tape, reflective; Nonexposed Lens.
O-E-915	-	Extinguisher, Fire, Dry-Chemical, (Portable).
O-D-1407	-	Dry Chemical, Fire Extinguishing, Potassium Bicarbonate.
W-B-131	-	Battery, Storage, Vehicular, Ignition, Lighting and Starting.
ZZ-T-381	-	Tire, Pneumatic, Vehicular (Highway).

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MIL-T-704	-	Treatment and Painting of Materiel.
MIL-G-5572	-	Gasoline, Aviation, Grades 80/87, 100/130, 115/145.
MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-F-8901 - Filter-Separators, Liquid Fuel: and Filter-Coalescer Elements, Fluid Pressures Inspection Requirements and Test Procedures for.
MIL-A-25896 - Adapter, Pressure Fuel Servicing, Aircraft, Nominal.
MIL-H-27516 - Hose and Hose Assembly, Nonmetallic, Suction and Discharge.
MIL-H-38209 - Hose Reel Assembly, Fuel Servicing, General Requirements for.
MIL-I-45208 - Inspection Systems Requirements.
MIL-F-52308 - Filter Element, Fluid Pressure.
MIL-M-81380 - Monitor, Contamination, Fuel Dispensing System.

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.
FED-STD-595 - Colors.

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-248 - Welding and Brazing Procedure and Performance Qualification.
MIL-STD-276 - Impregnation of Porous Nonferrous Metal Castings.
MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailer and Dollies: Preservation and Packaging of.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1223 - Non-Tactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification Marking and Data Plate Standards.
MS24484 - Adapter, Pressure Fuel Servicing, Aircraft, Nominal 2.5 Inch Diameter.
MS27031 - Coupling Half, Quick Disconnect, Cam and Locking Type, Drilling Template.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

Title 45 - Environmental Protection.
Part 1201 - Motor Vehicle Air Pollution Standards.

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Title 49 - Transportation.
Part 178 - Shipping Container Specifications.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

NATIONAL BUREAU OF STANDARDS (NBS)

H44 - Specifications, Tolerances, and Regulations for Commercial Weighing and Marking devices.

(Application for copies of CFR, DoT, and NBS publications should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Interstate Motor Carrier Noise Emission Standards.

(Application for copies should be addressed to the Public Affairs Office, Enviornmental Protection Agency, Rockville, MD 20852.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense (DoD) adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN PETROLEUM INSTITUTE (API)

RP1004 - Bottom Loading for MC306 Truck Vehicles.
RP2003 - Protection Against Ignitions Arising out of Static, Lighting, and Stray Currents.

(Application for copies should be addressed to the American Petroleum Institute, 2101 L Street Northwest, Washington, DC 20037.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B209 - Sheet and Plate, Aluminum and Aluminum Alloy.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section VIII.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.
No. 407 - Aircraft Fuel Servicing, Standard for.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA

California Vehicle Code.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The truck shall be a commercial product of manufacturer's latest design and shall meet the requirements specified herein. A single compartment aluminum tank, of 5,000-gallon capacity, shall be mounted on a diesel-engine-powered, six wheel, four rear wheel driven chassis with tilt cab or manufacturer's standard cab, of not less than 56,000 pounds (lb) gross vehicle weight (gvw). The truck shall be equipped with pumping and bottom loading system, filter/separator, fuel quality monitor, relaxation chamber, meter, and all components required to make a complete aircraft refueling system.

3.2 First article. When specified (see 6.2), the contractor shall furnish a truck for first article inspection and approval (see 4.2.1 and 6.4).

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3.2.1 Inspection system requirements. Quality assurance, inspection, and tests necessary to substantiate product conformance to contract specifications shall be in accordance with MIL-I-45208.

3.3 Standard truck, components, and accessories. Except as specified in 3.3.1 through 3.3.8, the truck, components, and accessories shall be standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented and rated in the manufacturer's sales information, including special or mounted equipment. Sales information shall be limited to specifications and technical material, identical to that furnished to the authorized company representative for selection of truck models and components. Trucks, components, assemblies, and accessories to be provided hereunder are standard or optional commercial products for the purpose of this specification. Such items shall be construed as commercial motor vehicle products and shall comply with all DoT Federal Motor vehicle Safety Standards and Regulations applicable, on the date of manufacture.

3.3.1 Drain plugs. Drain plugs installed in rear axles shall be of the permanent magnet type.

3.3.2 Heavy-duty cooling system. A heavy-duty cooling system shall be furnished that shall maintain engine coolant at a temperature below the boiling point with truck loaded to rated gvw, and operated at an altitute of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (°F).

3.3.3 Coolant system and indicators. Coolant system shall include a de-aeration system, a surge tank, or a coolant recovery reservoir of not less than 2 quart capacity. A low coolant level or high coolant temperature alarm buzzer and red indicator warning light shall be provided on the cab dash instrument panel. On tilt cab models, a radiator servicing access door shall be provided to allow verification of coolant level.

3.3.4 Towing devices. Not less than two hooks or loops for towing the truck shall be furnished on the front of the truck.

3.3.5 Radio noise suppressors. Electromagnetic radiation from the truck shall be within the limits of SAE J551.

3.3.6 Sound level. The truck exterior sound level shall conform to the EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Truck. The interior sound level shall be in accordance with the DoT Federal Motor Carrier Safety Regulations, section 393.94.

3.3.7 Air pollution control. Trucks, shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, trucks destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

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3.3.8 Lifting and tying down attachments. When specified (see 6.2), truck lifting and tying down attachments shall conform to type II or type III of MIL-STD-209 "Slinging Eyes and Attachments for Lifting and Tie Down Military Equipment." A non-ferrous transportation plate shall be provided and mechanically attached to the equipment. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tying down attachments may be identified by stenciling or other suitable marking. Tie down marking shall clearly indicate that the attachments are intended for the tie down of the equipment on the carrier when shipped.

3.3.9 Air transportability. When specified (see 6.2), the truck shall be air transportable by limiting the overall height to 104 inches and shall be designed to approach and depart a ramp at an angle of 17° (see paragraph 3.7.6 and 3.10.8).

3.4 Materials.

3.4.1 Protective treatment. Materials used in construction of the truck, that are subject to deterioration, shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with use, age, or extremes of climatic and environmental conditions shall be avoided.

3.4.2 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest practicable extent. Materials that are nutrients for fungi shall be treated with a fungicidal agent.

3.4.3 Bonding. Bond straps or bolted metal-to-metal conductive type connections utilizing cadmium plated external/internal lock washers shall be furnished between the chassis frame, and the cab, fenders, engine hood, and tanker body and also between pump, meter, filter/separator, fuel quality monitor, relaxation chamber, or other static producing devices and nearest structure part of the body or chassis. Bonding devices shall assure positive electrical connection at each, attaching point.

3.4.4 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service. Zinc, zinc-coated, copper, or copper-bearing alloys shall not be used in the filter/separator or piping downstream of the filter/separator (brass or bronze fittings may be used for hose fittings and nozzles). The metals that are used in the pumping system, bottom loading system, and all components which come into contact with the fuel shall be fabricated of nonferrous metals or 300 series stainless steel. Aluminum castings may be impregnated in accordance with MIL-STD-276 to prevent minor weeping.

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3.4.5 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimmilar metals are defined in MIL-STD-889.

3.4.6 Elastomers. Elastomeric materials used in the tank, pumping system, bottom loading system, and all components which come into contact with the fuel shall be compatible with fuels conforming to MIL-G-5572 and MIL-T-5624.

3.4.7 Anodizing. All aluminum moving parts and parts subject to wear shall be given an anodic coating.

3.5 Fonctional design. The truck shall be designed to satisfactorily receive, transport, store, and service all types of aircraft with fuels conforming to MIL-G-5572 and MIL-T-5624.

3.6 Identical items. All units of the same classification furnished under a specific contract shall be physically and mechanically identical. Parts, accessories, assemblies, and components are included in this requirement. Written approval for deviations must be obtained in advance from the contracting officer.

3.6.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2). In this specification, all measurements, dimensions, sizes and capacities are given in U.S. units.

3.7 Truck weights; ratings, and dimensions.

3.7.1 Curb weight. Curb weight shall include weight of chassis and cab, with all attachments, accessories, equipment, body, full complement of fuel, lubricants, and coolant.

3.7.2 Weight rating. The gvw rating shall consist of curb weight, operator (weight computed at 175 lb), and payload to provide not less than 56,000 lb gvw. In addition, the actual gvw of the truck furnished shall exceed the minimum gvw specified when required, to provide a payload of not less than 32,000 lb.

3.7.3 Weight distribution. The distribution of the gvw for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area.

3.7.4 Ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. Minimum gvw shall be 56,000 lb.

3.7.5 Wheel loading. The difference in load between the left and right wheels of any axle shall not exceed 2-1/2 percent of the load on the axle for any cargo load condition. The load on the left and right wheels of the bogie assembly may be measured in lieu of the wheels of the individual axles of the bogie assembly.

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3.7.6 Dimensions. The truck, without payload, resting on level ground, shall have the following dimensions:

a. Overall width, not more than 102 inches with all access openings closed.
b. Overall height shall not exceed 104 inches.
c. Overall length of the truck shall not exceed 34 feet.

3.7.7 Bogie axle spacing. The bogie shall have an axle spacing of more than 48 inches, but not more than 54 inches.

3.8 Performance.

3.8.1 Mobility. The truck shall be capable of transporting rated payload of 5,000 gallons of fuel, over highways or improved roads, at a speed of 45 miles per hour (mph).

3.8.1.1 High speed gradeability. Truck shall ascend continuous grades of 1.15 percent at 35 mph at rated gvw. Gradeability requirements shall be met with main transmission in direct drive. Gradeabilty shall be verified with calculations in accordance with SAE J688, truck ability prediction procedure (see 6.5).

3.8.1.2 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 45 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{governed speed (rpm)}}{\text{total gear reduction X tire factor}}$$

3.8.2 Service brakes. Service brakes shall control and hold truck when loaded to its specified gross weight, on a 30 percent grade. The service brakes shall stop the truck, loaded to specified gvw, within the stopping distance requirements of Federal Motor Carrier Safety Regulations, section 393.52.

3.8.3 Environmental conditions. The truck shall operate satisfactorily under the following conditions without adverse effects:

a. Temperature ranging from -20°F to +125°F.
b. Relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of water and frost.
c. Fungus growth as encountered in tropical and subtropical climates.
d. Salt atmosphere as encountered near bodies of salt water and in transportation on shipboard.
e. Sand and dust as encountered in normal and desert conditions.

3.9 Chassis components.

3.9.1 Chassis frame. The chassis frame shall be the manufacturer's standard, except as modified to comply with specific requirements specified herein. The truck shall have heavy-duty main frame or frame shall be

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reinforced, with the reinforcement extending at least from the rear of the front suspension rear hanger bracket to the bogie trunnion mounting bracket. The reinforcement shall provide sufficient structural strength in the chassis frame through increased resisting bending moment to at least equal the loads imposed with the truck loaded to provide the gvw specified herein. Heavy-duty frame shall have frame rails of greater section modulus than the manufacturer's standard for the gvw furnished and shall provide structural strength at least equivalent to the reinforced frame specified herein. Chassis frame rails shall not project beyond the end of the body.

3.9.2 Engine. The truck shall be powered with a diesel engine having the horsepower (hp), torque, and speed characteristics to satisfactorily meet all vehicular performance requirements specified herein on No. 2 diesel fuel. The engine shall be a liquid cooled, compression ignition, two-stroke or four-stroke cycle and may be naturally aspirated or turbocharged. Engine starting system shall include an ether starting system or glow plugs. Engine starting equipment shall include thermostatic controls of engine coolant temperature. Control of coolant temperature shall be accomplished by either thermostatically controlled radiator shutters or by thermostatically controlling the flow of all engine coolant through the radiator. Engine net hp figures used in performance predictions calculations (see 6.5) shall be determined in accordance with SAE J816 or J1349.

3.9.2.1 Oil filter. Full flow type oil filter shall be furnished.

3.9.2.2 Governor. Engine governor shall be furnished and set and sealed to limit engine-to-engine manufacturer's maximum recommended operating range.

3.9.3 Suspension. Truck shall be equipped with suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with truck loaded to rated gvw. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. Truck shall be equipped with hydraulic, double-acting, shock absorbers at the front wheels.

3.9.4 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground with truck loaded to rated gvw. A single reduction rear axle shall be furnished.

3.9.5 Rear bogie. The rear bogie on the vehicle shall be of the four-wheel type, complete with axles, springs, torque rods and all other necessary parts. The bogie shall be provided with means permitting differential action between the two axles, and a manually or automatically controlled lockout assuring equal power to each rear axle. Manual lockout control used shall be located in the truck cab. Axle gear ratios shall provide performance specified in 3.8.1.

3.9.6 Transmission. The truck shall incorporate a semiautomatic, continuous drive, automatically stepped transmission. The transmission assembly shall be of a commercial design and capacity that has been produced

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and supplied as a part of a commercial truck of comparable gvw rating. Transmission shall provide not less than four forward speed gear ratios plus one reverse speed, plus a variable conversion ration. The transmission shall be one in which a portion of the total torque conversion ratios are continuously variable, but in which the remaining ratio conversions are accomplished through range control by the use of automatically-stepped gear ratios that do not require more than an instantaneous power flow interruption with each selected range.

3.9.6.1 Selection of ranges of gears. The transmission shall provide for drive selection of ranges of gears required for efficient operations throughout all speeds to be encountered during operation of the truck. The shift control shall incorporate a positive indicator for each gear range selected. Drive gears for the power takeoff (pto) shall be provided (see 3.9.7). The transmission shall be so protected that the operator cannot inadvertently shift the truck into reverse.

3.9.6.2 Safety device. A safety device located at the transmission shall be installed to prevent starting of the engine in any selector setting other than neutral.

3.9.6.3 Torque capacity. The transmission shall have a torque capacity adequate to transmit 110 percent of the full throttle net engine torque, at any engine revolutions per minute (rpm), which could be encountered under any operating condition. A manually operated clutch member will not be acceptable.

3.9.6.4 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as delivered through the maximum gear train reduction. Driveshaft shall be protected in a manner similar to that specified for buses by DoT Federal Motor Carrier Safety Regulations, section 393.89.

3.9.7 Power takeoff. A split shaft or transmission side mount pto shall be furnished. The pto shall have sufficient capacity to drive the pump for the maximum load conditions to be encounted during operation of the truck. The pto drive shaft shall be heavy-duty type. Correct phasing and proper universal joint operating angles relative to maximum engine/pump rpm requirements shall be incorporated into installation design and assembly. The pto and pump gearbox gear ratios shall be selected to provide the pump operations at sufficient engine speed to assure adequate engine and gearbox lubrication and engine cooling.

3.9.7.1 Power takeoff control. The pto engagement and disengagement shall be controlled by the driver while sitting in the driver's seat. The control lever shall not interfere with normal operation or access to the driver's seat by the driver entering or exiting the cab. Other controls shall be installed as required, to provide a positive, smooth, mechanical engagement. Any engagement device which can cause clashing or mechanical parts or heavy shock loading will not be acceptable. The control mechanism shall be rigidly constructed to prevent damage and distortion over long periods of operation.

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3.9.7.2 Power takeoff safety devices. An interlock system shall be provided to prevent simultaneous propelling of the truck and operation of the pump. An interlock system shall be provided to prevent shifting of the pto control in either direction unless the engine is at idle speed or unless the service brakes are set. A safety device shall be provided to prevent operation of the throttle control, when the pto is in the truck drive position.

3.9.8 Fuel system. Fuel system shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.65 and 393.67.

3.9.8.1 Air cleaner(s). Air cleaner(s) shall be provided. Air cleaner(s) shall be as recommended for all-climate operations including heavy dust and sand (see 3.8.3). When equipped with an oil bath type, the oil capacity of each air cleaner shall be not less than 2 quarts.

3.9.8.2 Fuel tank. Fuel tank shall be manufacturer's standard for the truck furnished. Minimum capacity shall be not less than 43 gallons. When more than one fuel tank is furnished, means shall be provided to assure equalized fuel level in both tanks.

* 3.9.9 Exhaust systems. The truck's exhaust system shall conform to paragraphs 4-3.4 through 4-3.4.5 of NFPA No. 407 (84 edition). Particular attention in design and location shall provide for a cover plate of exhaust system portions that project to the rear of the cab. The cover plate shall form a shield that will shed fluids, and preclude spilled fuel from coming into contact with the exhaust system. The outlet discharge shall be directed away from the pump and pumping system compartment. A spark arrester shall be furnished except on turbo charged engines. The spark arrest shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

3.9.10 Electrical equipment.

3.9.10.1 Starting system. The truck shall be furnished with a 12- or 24-volt (V) starting system, with a 12 V lighting system, and shall be provided with an alternator of not less than 60 ampere rating capacity.

3.9.10.2 Lighting and wiring. The truck lighting and wiring shall conform to and be installed in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.12, 393.19, 393.20, 393.22 through 393.26(d), 393.27 through 393.31, 393.33, and as specified herein. Wiring, lamps, and all electrical equipment and components installed in the hazardous vapor area is shall conform to the following requirments: The hazardous vapor area is identified as the entire rear portion of the truck from back of the cab. Wiring for lamps shall be not smaller than 14 gage, stranded, with thermoplastic insulation, with the exception of lamp pigtails which may be 16 gage but not longer than 6 inches.

3.9.10.3 Wiring system installation. The wiring system components and installation in hazardous vapor areas defined in 3.9.10.2 shall comply with Article 501 of NFPA No. 70 for class I, division 2 locations.

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3.9.10.4 Battery(s). The battery(s) provided with the truck shall have sufficient capacity to meet the starting and operating requirements specified in 3.9.10.1. Each battery shall be a 12V battery. The total reserve capacity rating shall be not less that 320 minutes and the total cold cranking rating of 0°F shall be not less than 1200 amperes, for engines with not less than 220 gross horsepower and 1800 amperes for engines with 220 gross horsepower or more. Reserve capacity and cold cranking ratings shall be in accordance with SAE J537. The battery(s) shall not be located in the cab and shall be installed in an accessible area. The battery cables shall be protected by use of grommets when the cables pass through metal bulkheads. There shall be sufficient clearances between battery terminals and battery storage containers when connecting cables are in place. The installation of the battery(s) shall provide for each access for checking the electrolyte without removing the battery(s). The battery(s) shall be of the maintenance free type having the maintenance free characteristics listed in W-B-131.

3.9.10.5 Lamps. All lamps installed in the hazardous vapor area shall be of permanent explosion proof design. The lamp body shall be nonferrous and shall incorporate a screw-type fitting with gasket for making a vaportight connection with the wiring conduit The lens shall be heavy duty and capable of normal removal and installation without cracking or breaking. The lens shall be designed to provide a vaportight seal with the lamp body. The lamp assembly shall be designed to withstand a 3 lb per square inch (psi) positive or negative pressure without leaking. The vaportight seal shall not be attained by the use of cements, sealant, or similar materials. The lamp bulbs shall be of the rough-service type. Lamps shall comply with paragraph 4-3.8.6 of NFPA No. 407 (84 edition).

3.9.10.6 Protective devices. Overload circuit breakers shall be used in the hazardous vapor area in lieu of replaceable, fuse-type devices. Circuit breakers shall be of the explosionproof design. All circuits shall be protected by circuit breakers. Protective devices shall be connected and identified in an organized manner to facilitate determination of malfunction.

3.9.11 Wheels, rims, tires, and tubes. Truck shall be equipped with single front and dual rear wheels. Rim and tire ratings shall conform to TRA recommendations, for the type and size of tires furnished. Tire and rim sizes and tire ply rating shall be the same for all wheels on each truck. All wheels shall be disk type.

3.9.11.1 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with truck loaded to rated gvw. Tires shall conform to TRA or to ZZ-T-381 with a size designation system the same as the TRA.

3.9.11.2 Inner tubes. For tube tires, inner tubes shall be heavy-duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with TRA recommendations.

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3.9.11.3 Tire chain clearance. Clearance shall be provided for installation and use of tire chains on rear wheels in accordance with SAE J683.

3.9.11.4 Spare wheel. When specified (see 6.2), an inflated tire is mounted on a spare wheel shall be furnished and shall be stowed and secured on the truck for shipment.

3.9.12 Brakes. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.40 through 393.43 and 393.45 through 393.52, as applicable.

3.9.12.1 Service brakes. Truck shall be equipped with full air brakes on all wheels. The braking system, complete with all necessary components, shall include:

a. Air compressor, unloader head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute.
b. Air storage reservoir(s) with not less than 3,000 cubic inches total capacity; each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank.
c. Foot control, suspended or treadle type.
d. Air pressure gage, visible to the driver.
e. Low-air-pressure warning, visible and audible.
f. Air control valves.
g. Service brake stop lamp switch.
h. An air dryer provided with an automatic moisture ejector and heating device.

3.9.12.2 Parking brakes. Truck shall be equipped with parking brake capable of being applied and maintained while engine is running.

3.9.13 Rear bumper. A full width, heavy-duty, rear bumper shall be mounted on the chassis frame. The bumper shall be constructed of steel channel section. The outer corners of the bumper shall be rounded and shall incorporate other protection as required to minimize the hazard to personnel walking around the truck. The bumper shall be so designed that the fully loaded truck can be pushed without damage.

3.9.14 Fenders. Fenders shall be provided over the rear wheels of the truck. The fenders shall be of heavy-gage steel and supported to withstand a static load of 400 lb placed on top of the fenders. The outer edges of the fenders shall be rolled or reinforced with rolled steel angle. Skirting may be supplied in lieu of fenders, provided it is designed and constructed to provide the required protection.

3.9.15 Mud flaps. Substantial, flexible mud flaps shall be mounted behind the front wheels and the bogie, and in front of the bogie. The ground clearance of the mud flaps shall be not more than one-third of the distance from the center of the closest axle to the center of the flaps, under any load

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condition. The mud flaps shall be mounted within 12 inches of the tires. The flaps shall be at least as wide as the tires. The flaps in front of the bogie shall be supported so as not to interfere with the running gear. If the truck is constructed so that these requirements are accomplished by means of fenders or body construction, mud flaps shall not be required.

3.9.16 Steering. Manufacturer's standard power steering shall be furnished.

3.9.17 Cab. Manufacturer's standard full width or tilt cab shall be furnished with the tilt cab having tilting and locking mechanisms. Locking mechanism levers shall be readily accessible to the operator while standing on the ground. Cab doors shall be equipped with locks, operable from inside the cab, and with at least the curbside door equipped with external, key operated lock. Cab shall be all steel type with upholstered, full-width, adjustable seat or individual seats, the driver's seat being adjustable. The doors shall open wide, and safety grips shall be furnished on each side of cab, to assist personnel wearing heavy arctic clothing to enter and leave cab. The driver's door shall be equipped with a device to hold the door open when the vehicle is stationary and pumping product fuel, to enable the driver to re-enter the cab during emergency drive-away conditions. For tilt cab, a grab handle under the windshield and a bumper step, or bumper step cutouts, shall be provided to facilitate cleaning the windshield. Interior lighting shall be provided. Fastenings and two pair of seat belts shall be installed.

3.9.18 Windshield wipers and washers. Truck shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by electric motor(s) or air.

3.9.19 Heater and defroster. Hot water heater shall be provided. Heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to floor and to defroster louvers. Heater shall be complete with blower and mounted controls convenient to the driver.

3.9.20 Controls and operating mechanisms. All controls and operating mechanism shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing.

3.9.21 Tools. When specified (see 6.2), each truck shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of loaded truck to a height adequate to permit removal and replacement of wheel and tire assembly.

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3.9.22 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Combination (flat and convex) mirrors enclosed in a common housing and each with at least three supporting arms shall be mounted on each side of the cab. Each combination mirror shall have at least 50 square inches of flat reflective area and convex surface having at least 20 square inches of reflective area.
b. Key-operated ignition switch.
c. Ammeter or charging indicator.
d. Fuel gage.
e. Oil pressure gage or indicator.
f. Engine temperature gage or indicator.
g. Speedometer with recording odometer.
h. Dual sunvisors.
i. Armrest on driver's side.
j. Driver's compartment ventilator other than window.
k. Electric or air horn.
l. Tachometer.
m. Engine hour meter totalizing not less than 9,999 hours.

3.10 Tank. The tank shall be a cargo tank for transporting aviation gasolines or jet fuels. The tank shall be of the single compartment type, with a capacity of not less than 5,000 gallons plus at least 3 percent expansion space. The tank and end plates shall be constructed of 5,000 series alloy sheet aluminum of sufficient thickness to provide structural integrity of the tank for the life of the vehicle. The aluminum base metal shall have the composition, physical, and mechanical properties and relative weldability characterisitics, equal to ASTM B209, 5086-H32, 5086-H36, or 5456-H321 aluminum alloy. The tank shall be elliptical or modified rectangular cross section. The tank interior shall have a bright finish and no evidence of water-stained aluminum will be permitted. Adequate hold-down devices shall be provided to anchor the product fuel tank in a suitable manner that will not introduce undue concentration of stresses, and shall be built to withstand loadings in any direction equal to the weight of the tank and attachments when filled with product fuel. The complete tank design and construction shall conform in all respects to the requirements of CFR 49, paragraphs 178.340 and 178.341 of MC306. All welded joints of the tank shell, tank end plates, and baffles shall be of the inert Gas-Arc welding process. Tank shell fabrication and assembly shall be by longitudinal automatic continuous weld seams. The tank shell assembly shall have not more than one circumferential weld seam other than at front and rear end plates. The tank shall be equipped with a fully automatic bottom loading and vent system, and emergency type flow control valve for product discharge. Extrusions, pipe, tubing, and fittings used in assembly of the tank shall possess characteristics adequate for operational requirements. Tank brackets and structural supports shall have an alloy composition equal to ASTM B209, 6061 aluminum alloy. At the manufacturer's option, the tank shell, baffels, bulkheads, pads, catwalk, manholes, and outlets shall be of 12 gage (0.1046 inch) type 304 stainless steel (or heavier when needed to comply with the maximum allowable stress

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level of DoD Safety Regulation 178.340). The stainless steel tank brackets and structural supports shall be type 304 stainless steel. The vehicle shall comply with DoT Safety Regulation 178.341, of MC306 and all supplements to DoT Regulations applicable to MC306.

3.10.1 Static pressure. The tank shall withstand an internal static air pressure of at least 3 pounds per square inch gage (psig) without evidence of leaking.

3.10.2 Baffles. Baffles shall be provided. The baffles shall be designed and constructed in accordance with CFR 49, paragraph 178.340 of MC306 and shall permit inspection and cleaning of the entire tank. Baffles shall be adequately reinforced and welded to the tank shell. Either flanged or nonflanged design may be used on the outer edge of the baffle. If nonflanged design is used, both sides of the baffle shall be welded to the shell at not less than 66 percent of the perimeter and each head and baffle shall be adequately braced with structural members. Each baffle shall also be provided with a top and bottom half round opening with not less than 9-inch radii, and a manway opening of not less than 18 inches in diameter. The 18-inch openings shall be staggered to one another to resist surging of the product fuel. Flanges for heads and baffles shall be at least one inch in width around their outer periphery and radius of bend shall be adequate to relieve stress concentrations at junction with outer shell.

3.10.3 Tank calibration marker. A fixed marker, visible through the manhole, shall be provided inside the tank to indicate when the tank is filled with 5,000 gallons of fuel.

* 3.10.3.1 Fuel load chart. A temperature compensated chart for converting inches of depth in the tank to gallons of fuel in the tank shall be provided. The chart shall be permanently installed so as to be readily visible. The chart shall be fabricated of a material so as to remain legible for the life of the vehicle.

* 3.10.4 Manhole. The tank shall have a circular manhole, with cover, with a minimum diameter of 20 inches. The manhole cover shall have a 10-inch, watertight, hinged, filler cover with necessary safety features to eliminate the possiblity of a spark occurring during opening or closing of the cover. The manhole shall be equipped with a buna-N gasket, or equivalent, to insure a fuel-tight seal between the collar and the cover. The entire manhole cover shall be removable for entrance into the tank by removing a bolt-on clamping ring. Safety devices to prevent the manhole or fill cover from opening fully when internal pressure is present shall be provided. The cover shall have a forward-mounted hinge and self-latching catches. A 2-inch National Pipe Thread coupling shall be provided near manhole cover in the tank top to permit installation of fuel level sensing device as described in paragraph 3.14.3.

3.10.4.1 Tank opening protection. All tank filler openings and vents shall be protected against overturn damage by a rigid member or members firmly fixed to the tank. Specific design of overturn protection devices shall be in accordance with CFR 49, paragraph 178.340-8 of MC306.

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3.10.5 Vents. Tank vents shall be provided with safety devices to prevent excessive pressure from building up under both normal and emergency conditions. Vents shall be designed in accordance with CFR 49, paragraph 178.341-4 of MC306. For normal venting under loading and unloading conditions with dome cover closed, pressure-vacuum vents shall limit the vacuum of 1 psi and tank pressure of 3 psig based on maximum product transfer rate. Emergency venting can be obtained by pressure activated and fusible type vents. Fusible vents and pressure-vacuum relief valves can be components of the manhole assembly or used separately. The vent shall be designed to avoid the possibility of malfunction or jamming and shall not contain metal alloys which could corrode and cause malfunctions. There shall be no steady stream leakage of fuel under full tank head pressure should the truck upset with the vent position from vertical of up to 100° in either direction.

3.10.5.1 Vapor recovery system. When specified (see 6.2), the truck shall be equipped for vapor recovery in general conformance with the recommended practice of the American Petroleum Institute (API) in API RP1004, as summarized herein. Vent valves shall be furnished with an interlock to assure that the vent is open when bottom loading shutoff valve is open. The vent valves shall be furnished with a vapor recovery hood or collecting chamber and manifolded to the rollover rail or other suitable manifold piping. The manifold piping shall terminate at the rear of the truck or near the bottom loading stub and at that point shall be provided with a standard 4-inch cam-and groove type quick coupler adapter designed to comply with API RP1004. Connections between the tank vapor space vents and the header tube(s) shall be not less than 7 square inches in open cross sectional area. The header tube shall be not less than 7 square inches in open cross sectional area. A dust cap with chain is to be provided with the coupler. The vapor recovery system shall be designed to prevent pressure buildup during loading or an excessive vacuum condition during off loading. Proper size vents in the vapor recovery system are required to protect the truck in the event the vapor line is closed for any reason. Provision shall be made for discharging to atmosphere the vapors from the vapor recovery system when facilities are not available for vapor collection. The location for the outlet shall be on the top of the vehicle, remote from the driver's loading area, and isolated from the engine exhaust.

3.10.6 Catwalk. A catwalk shall be provided and shall be integral with and securely attached to the tank. The catwalk shall be constructed of embossed diamond type, expanded metal safety tread, or equivalent. The catwalk shall span the manhole, extend the length of the tank, and incorporate an adequate landing platform to accommodate the boarding ladder specified in 3.10.7. The catwalk shall be not less than 30 inches wide. Means shall be provided for draining entrapped water from the catwalk. The catwalk shall be so designed that fuel will not run off the front of the tank in the event the tank is overfilled and fuel is expelled from the manhole or vent. Tubular or formed scuff guards shall be of sufficient height and shall incorporate structural integrity to protect the manhole and other appurtenances mounted on top of the tank in the event of overturn.

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3.10.7 Boarding ladder. A tubular steel or tubular aluminum ladder shall be securely attached to the rear of the tank to provide access to the catwalk. The ladder shall be mounted in the plane described by the rearmost surface of the bumper and in the center of the tank. The ladder shall be constructed and mounted to provide a rigid structure when scaled by a 220-lb man. A rigid step shall be suspended on the underside of the bumper to facilitate access to the ladder. The vertical members of the ladder shall be 18 to 20 inches apart. The ladder height shall be the same as the height of the tank. Demountable grab handles to extend the ladder upward 12 inches and forward 18 inches shall be provided. The ladder assembly shall be inherently safe and convenient for operating personnel.

* 3.11 Pumping system compartment. Pumping system controls, dispensing reel and hose, meter, static reel, bottom loading and defuel inlet manifolds and pumping system related piping and valves shall be housed in an open compartment, accessible for normal operations from the left or driver's side of the truck. The compartment assembly shall be located under the product fuel tank and shall incorporate necessary structure integrity to prevent binding, warpage, or other types of damage due to vibration and other forces encountered during normal operation of the truck and pumping equipment. The floor shall be designed to insure that spilled fuel will not accumulate in the compartment. Lights shall be provided to fully illuminate the interior of the compartment and allow observation of all controls, instruments, and instruction plates for night operations. The lights used in this area shall be of the type approved for use in hazardous locations as described in class I, division 2 of NFPA No. 70.

3.12 Pumping system. The pumping system shall include the tank-to-pump suction line, pump, filter/separator, fuel quality monitor, relaxation chamber, meter, dispensing hose reel, dispensing hose, nozzles, system bypass line, control system, control panel, and defuel system. The complete pumping system specified in 3.12.1 through 3.12.19 shall be designed to handle aviation gasolines conforming to MIL-G-5572 and turbine (jet) fuel, grades JP-4 and JP-5 conforming to MIL-T-5624; and shall provide the performance specified herein when handling these fuels. Operation of the pumping system for fueling and defueling aircraft shall not involve the use of external sensing hoses, or other devices on any other equipment other than that specifically referred to herein. There shall be no electrical equipment or devices associated with or incorporated as a part of the pumping system except as specifically referenced herein. The pumping system shall operate for continuous periods of several hours without degrading effects.

3.12.1 Hydrostatic pressure. The piping system and all components shall withstand a hydrostatic pressure 1-1/2 times the maximum pressure encounted during operation of the truck. Piping shall be constructed of ASTM B209 aluminum alloy 6061, temper T4 or T6 or schedule five stainless steel, or equal.

3.12.2 Pump. The pump shall be of the self-priming, centrifugal, aircraft, fuel dispensing type driven directly by the pto. The pump and pumping system shall provide the performance specified herein when handling

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fuels conforming to MIL-G-5572 and MIL-T-5624. The pump shall be capable of producing a vacuum of not less than 18 inches mercury (Hg) when loading the truck from an outside source or defueling aircraft.

3.12.2.1 Pump seal failure. The pump shall be so constructed and positioned that fuel will not enter into any gearbox or transmission in the event of pump seal failure.

3.12.2.2 Pump drain. A drain plug or valve shall be provided for completely draining the pump.

3.12.3 Pumping system operation. The pumping system shall be designed for a servicing rate of 300 gallons per minute (gpm). The pumping system shall exhibit no instability in pressure or flow rate under any operating condition The pumping system shall function as follows:

a. Pump from the tank through the filter/separator, fuel quality monitor, relaxation chamber, meter, 2 inch servicing hose, and overwing nozzle at a flow rate adjustable from zero to up to but not more that 100 gpm. Once the pumping system controls have been adjusted, flow shall be regulated on demand by control of the overwing nozzle.
b. Pump from the tank through the filter/separator, fuel quality monitor, relaxation chamber, meter, an out through the 2-inch servicing hose at a minimum flow rate of 250 gpm at a minimum pressure of 30 psig, and a maximum pressure of 55 psig, measured at the nozzle sampling port. Once the pumping system controls have been adjusted, flow shall be regulated on demand ahead of the underwing nozzle. The system shall automatically recover from an externally induced no-flow condition to full flow within 15 seconds after the full-flow demand condition has been re-established.
c. During underwing operation, the pumping system shall limit the total pressure at the inlet of the underwing nozzle to 120 psig when transcending from maximum flow to shutoff in 0.5 seconds. This capability shall be measured with the hose and regulator remove.
d. In the defueling mode, pump through the defuel system at a valve regulated flow rate of 0 to not less than 80 gpm, but not exceeding 100 gpm.

3.12.4 Tank-to-pump suction line. The tank-to-pump suction line shall be designed to provide adequate flow and smooth operation of the pump under all conditions specified herein. The suction line shall incorporate a line strainer and tank shutoff valve. A flexibe coupling may be provided, if required, by the design of the tank and compartment mounting. The suction line shall terminate in the bottom of the tank, and shall not have a raised seat that will allow sediment and water to collect in the low point area. The product discharge opening shall be of the emergency flow control type designed in accordance with CFR 49, paragraph 178.341-5 of MC 306 to assure against the accidental escape of contents. Internal valve shall be in accordance with 3.12.4.2. A tank drain shall be provided in the low point area, and shall be readily accessible.

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3.12.4.1 Line strainer. The suction line shall incorporate a nonferrous type 40 mesh strainer. The strainer shall be rigidly constructed and adequately sized to allow a 300 gpm flow at minimum pressure drop. The screen shall be easily removable for cleaning and inspection.

* 3.12.4.2 Internal valve. An internal valve shall be provided in the bottom of the product fuel tank in the lowest point to allow for water drain off. The internal valve shall serve as a combination fuel issue, bottom loading and emergency shut-off valve for all fuel dispensing and loading operations. The internal valve (emergency valve) shall be in accordance with design requirements of 49 CFR, Section 178.341-5 of MC306. A deflector, if required, shall be mounted in the tank area over the internal valve to prevent the fuel from spraying unrestricted into the tank. A 3/4 inch drain valve located on the drain line shall be provided in the outside piping down stream of the internal valve and shall be easily accessible to the operator from the outside of the truck for periodic fuel checking and draining.

* 3.12.4.2.1 Emergency shut-off system. The emergency (internal) valve (see paragraph 3.12.4.2) shall have emergency shut-off devices incorporated into the design that will allow for vehicle protection in the case of fire or malfunction of fuel dispensing systems. The valve shall be in a closed position at all times except when product is actually being pumped. A fusible link shall be incorporated in the emergency valve or its controls to provide automatic closure in the event of fire. The remote controls for the emergency (internal) valve may be of any accepted commercial design, either air or lubricated cable operated. The control shall be located at the forward left side of the tank, convenient to the operator. An emergency shut-off control shall be located at the rear right side of the tank. A sign marked "EMERGENCY SHUT-OFF" with letters at least three inches high shall be painted adjacent to this control."

3.12.4.3 Flexible couplings. If required to provide relative motion between the pumping system and the tank, flexible couplings shall be provided in the system bypass line and the system tank return line. The flexible couplings shall withstand an integral pressure of 150 percent of design pressure, and a vacuum of 0 to 15 inches Hg without leakage or other type failure. All pipe openings into the compartment shall be protected to limit air passage around the pipes into or out of the compartment.

3.12.5 Filter/separator. A filter/separator with a rated capacity of 300 gmp shall be provided. The filter/separator shall conform to MIL-F-8901, and shall incorporate filter elements conforming to MIL-F-52308. A sufficient number of canister assemblies shall be provided having seals and monel wire screen coated with polytetrafluorethylene for water separation. Each canister shall be capable of retaining 0.5 inch of water. Not less than 15 elements shall be provided. Instruction tags are to be attached at pumping control console noting that the filter/separator elements are new and unused. The filter/separator shall be so installed in the system that all fuel dispensed to aircraft, under all operating conditions, will pass through the filter/separator. Fuel shall pass through the filter/separator in one direction only, under any condition.

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3.12.5.1 Pressure vessel. The pressure vessel shall be designed and fabricated in accordance with MIL-F-8901 and ASME Boiler and Pressure Vessel Code, section VIII. The pressure vessel shall be designed for a working pressure of the maximum pressure encountered during operation of the pumping system. The pressure vessel shall be hydrostatically tested at 1-1/2 times the working pressure. The pressure vessel flange gasket shall be of the O-ring type. When the filter assembly is mounted vertically and it is necessary to remove the assembly for cleaning and servicing. The main vessel shall be provided with lifting devices capable of lifting the entire filter/separator assembly. A non-ferrous plate shall be affixed to the tank indicating contractors compliance to the ASME code.

3.12.5.2 Air eliminator. An automatic air eliminator of the standard commercial float activating type shall be provided to allow automatic purging of air from the filter/separator. The air eliminator shall be vented into bottom portion of the cargo tank. The air eliminator shall be of the manufacturer's recommended size to allow proper air elimination but shall be not less than 3/4 inch. A check valve shall be provided in the air eliminator line to prohibit the back flow of product fuel.

3.12.5.3 Water slug control. A water slug control shall be provided. The water slug control shall stop fuel flow when a predetermined water level is reached in the vessel and shall then permit flow when the water level is lowered to a predetermined level.

3.12.5.3.1 Automatic water drain. The water slug control shall contain an automatic water drain to automatically discharge water as it accumulates without interfering with the normal function of the pumping system.

3.12.5.4 Manual drain. The sump of the filter/separator shall be equipped with a manual drain of at least 1-inch nominal size.

3.12.5.5 Filter/separator pressure gage. The filter/separator shall be equipped with a pressure gage having a diameter of not less the 4-1/2 inches, graduated in 2 psi increments throughout the range of 0 to 160 psi. The gage shall be constructed of either nonferrous material or stainless steel. The gage shall have a recalibration feature. A means of preventing moisture accumulation on the inside of the gage glass shall be provided. The gage shall be mounted on the control panel. The gage shall be connected by a three-way valve and necessary tubing to provide measurement of the pressure differential across both the first and second stage of the filter/separator. A direct reading type differential gage is acceptable in lieu of three-way type providing the gage is graduated in 2 psi increments. When a direct reading gage is furnished it shall be of not less than 2-1/2 inches in diameter, shall have a range of 0 to 25 psi and shall have red and green areas indicating safe operation.

3.12.5.6 Filter/separator performance. The filter/separator performance shall be in accordance with MIL-F-8901 and as specified herein, with the following additions and exceptions.

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3.12.5.6.1 Differential pressure. The maximum allowable differential pressure in accordance with MIL-F-8901 shall be 10 psi in lieu of 5 psi.

3.12.5.6.2 Water separator. The 10 percent by volume water separation requirements of MIL-F-8901 is not required for this application.

3.12.6 Fuel quality monitor. A fuel quality monitor, meeting all requirements of MIL-A-81380, shall be furnished and installed downstream of the filter/separator. The fuel monitor shall contain a built-in surge suppressor that will provide maximum protection to the filter/separator in the event of a pressure surge in the system. One set of fuel monitor elements shall be provided. Instruction tags are to be attached at the pumping control console noting that the elements in the fuel quality monitor are new and unused. A pressure gage, identical to the gage specified in 3.12.5.5, shall be mounted on the control panel and connected by a two-way valve and necessary tubing to measure pressure drop. A direct reading type differential gage is acceptable in lieu of three-way valve type, providing the gage is graduated in 2 psi increments. Fuel monitor pressure gage is not required if fuel quality monitor is incorporated in the filter/separator.

3.12.6.1 Relaxation chamber. An aluminum relaxation tank of approximately 145-150 gallon capacity shall be provided and installed downstream of the filter/separator and fuel quality monitor. The tank shall be contructed to withstand at least 150 lb working pressure. The tank shall be baffled to insure complete turnover of product fuel and tank inlet and outlet shall be parallel to each other as far apart as possible. The piping to and from the tank shall be 3-inch aluminum, schedule 40, with all necessary bends and grooved fittings. A manual drain valve shall be provided in the bottom of the relaxation tank. Metal tubing of not less than 3/4 inch shall be provided as an air elimination system through constant flow of product fuel from the top of the relaxation tank into the bottom of the product fuel tank. The air eliminator tap shall be located in an area away from the entering fuel and shall terminate into a check valve in the product fuel tank that shall prohibit the back flow of fuel. The relaxation tank shall allow dissipation of static electrical charges generated in the product fuel during filtration and pumping. A charge relaxation time of 30 seconds shall be provided in accordance with API RP 2003. The relaxation tank shall be designed and constructed in accordance with ASME Boiler and Pressure Vessel Code, Section VIII, Division 1. A non-ferrous plate shall be affixed to the tank indicating contractor's compliance to the ASME code.

3.12.7 Meter. One positive displacement meter with maximum rating of at least 300 gpm shall be housed in the pumping system compartment. Meter shall be capable of momentary overspeeding to 125 percent of maximum rated capacity without mechanical damage or impairment of accuracy. Meter shall be designed for 75 psi working pressure and 125 psi hydrostatic test pressure. Meter shall have an accuracy in accordance with the requirements of NBS H44. Meter shall be capable of being adjusted while under pressure without leakage or loss of the product. The adjustment sensitivity shall be sufficiently fine to permit calibration changes in conformance to the accuracy requirements set forth above and shall be capable of an adjustment range so that meter shall be

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above and shall be capable of an adjustment range so that meter shall be suitable for aviation gasoline, JP-4 and JP-5 fuel, and shall be accomplished without the use of external change gears. A vertical meter counter shall have a seven figure totalizer and a four or five figure setback indicating full gallons. Figures in set back shall be coated with light reflecting material. All components of the meter in contact with the fuel shall be fabricated from aluminum material. Stainless steel may be used in conjunction with aluminum for moving components of the meter having metal-to-metal contact.

* 3.12.8 Hose reel and hose. A manually driven dispensing hose reel that is to be interchangeable for use in underwing and overwing fuel dispensing shall be provided in the pumping system. The hose reel shall conform to applicable paragraphs of MIL-H-38209 and as specified herein and shall be mounted in the forward underslung pumping system compartment at the driver's side of the truck. The hose reel shall be provided with 50 feet of 2 inch diameter hose as specified in MIL-H-27516D with internal bonding/tension wire. The hose ends shall be attached to adapter/couplings by stainless steel type banding. This 50 foot length of hose is to be used in overwing and underwing fuel dispensing and shall be made up as a special hose assembly kit equal to AERO QUIP AE706991-1 containing a hose end regulator (45 psi) on the nozzle end and a 2 inch NPT coupler on the other end for attachment to the emergency dry break connector described in paragraph 3.12.8.1. The nozzles and adapters are futher described in paragraph 3.12.8.2. Gear ratios (ring gear and pinion) shall be designed for easy manual rewinding. Provisions shall be made for a properly stowed and secured nozzle(s) and removable crank handle when not in use.

* 3.12.8.1 Emergency dry break connector. An emergency dry break connector of a straight pull, pressure insensitive type shall be installed at the fuel dispensing reel. This dry break connector is to permit the truck to drive away in the event of fire, breaking the hose at the reel, and leaving the hose and nozzle connected to the aircraft. Both halves of the connector are to seal instantly on breakaway eliminating fuel spillage from either truck or delivery hose. The connector breaking force shall be 275 lb (within a permissible range of a -25 lb and +50 lb). The reel shall also be designed to accept the dry break connector by providing at the reel a special section of hose (approximately 2 feet in length) with the dry break connector attached The hose shall be in accordance with MIL-H-27516D with internal bonding/tension wire to obtain electrical continuity with the fuel dispensing components. Both hose ends shall be banded to the 2 inch NPT male and female couplers by stainless steel bands. The hose reel drum is to be sufficiently recessed to accommodate the special section of hose and coupling so as to permit level winding for subsequent layers of hose. The 50 feet of 2 inch dispensing hose, as specified in 3.12.8, shall be banded and coupled (2 inch NPT, male) to the emergency dry break connector and the 2 foot special hose section.

3.12.8.2 Delivery nozzles.

* 3.12.8.2.1 Fuel delivery nozzles. The fuel shall be dispensed through overwing and underwing nozzles that are interchangeable on the special hose assembly described in paragraph 3.12.8 depending on the fueling mode desired.

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The overwing nozzle assembly and adapter shall be equal to AERO QUIP AE83956R. The underwing nozzle assembly and adapter shall be equal to AERO QUIP model RDF, AE83277R with a 60 mesh strainer, and fuel sampling part that is to contain a Gammon Technical Products, Inc., sampler (model GPT 235-1/4 inch). These nozzles shall incorporate a special AERO QUIP UNISEX nozzle adapter to alow for interconnection to hose assembly. A separate UNISEX adapter (AE82100R) shall be provided for use if the field activity required the use of a D-1 type nozzle. The nozzle accessories and the D-1 type nozzle adapter shall be securely mounted in the pump system compartment adjacent to the hose reel.

3.12.8.3 Hose reel shutoff valves. A shutoff valve shall be located upstream of the hose reel. The shutoff valve shall be hydraulically or pneumatically controlled by individual controls located on the pumping system control panel. The valve shall open when the control is placed in the ON position and the hoses are unreeled, and shall remain closed (pressure-tight) at all other times, including when the pump is not operating.

3.12.9 Bypass line. The bypass line shall relieve the pump under low-flow and no-flow conditions. An automatic bypass valve, which integrates with the control system, shall be provided to achieve the performance required herein. The bypass line shall be located upstream of the filter/separator. A pump bypass manifold installation shall be provided to prevent tank overfill fuel spill from occurring during defuel (tank filling) operation. The bypass shall prevent engine speed changes of more than 350 rpm between full-flow and no-flow for any pumping mode, except defuel. All bypass pilot valve shall be inactivated while defueling, with the exception of the maximum pressure pilot specified in 3.12.15 in order to prevent any possiblity bypassing fuel.

3.12.10 Check valves. If required, the system bypass line and the system tank return line, as well as any other piping or tubing between the tank and pumping system except the tank-to-pump suction line, shall incorporate check valves or manually controlled valved to prevent backflow from the tank when maintenance is performed on the pumping system. The check valves shall comply with CFR 49, paragraph 178.340-8 of MC306. Check valves or manually controlled valves shall also be located at the filter/separator and the fuel quality monitor to allow routine servicing and replacement of filter/separator elements and fuel quality monitor elements without draining adjacent piping.

3.12.11 Control system. The control system shall be hydraulically or pneumatically operated. The system shall automatically provide the pressure control and flow relief functions to accomplish the performance requirements specified herein. The control system shall prevent pump speed surges, pressure surges, and any other type of pumping system instability. The system shall automatically regulate to the required flow conditions, as determined by the overwing nozzle and the aircraft fuel system ahead of the pressure servicing nozzle. The control system shall include the system bypass valve, dispensing hose reel shutoff valves, filter/separator water discharge slug control, defuel control valves, and opeating controls.

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3.12.12 Selector valve(s). The selector valve(s) shall be the primary operating control of the pumping system. The selector valve(s) shall have positions for overwing, pressure servicing, and defuel. The selector valve(s) shall provide for automatic adjustment of the pumping system controls to accomplish the the corresponding pumping mode.

3.12.13 Control system maintenance adjustment. The control system shall be so designed that all control adjustment can be accomplished in a specified sequence, and all adjustments shall be independent. The control system shall be so designed that all adjustment points shall be accessible without the removal of any part of components.

3.12.14 Safety provisions. The pumping system shall be so designed that all valves and controls can be adjusted in any position without causing damage to any portion of the truck under any operating condition, or exceeding 50 psig at the pressure servicing nozzle during pressure servicing, or 300 gpm through the meter. Means shall be provided which will not allow the normal maximum pumping system operating pressure to be exceeded by more than 10 psi under any condition. All product discharge piping and inlet piping to the tank shall be in accordance with CFR 49, paragraph 178.340-8 of MC306 for accident damage protection.

* 3.12.15 Control panel(s). The pumping system shall incorporate control panel(s) on the left or driver's side of the truck in the front pumping compartment. The control panel(s) shall be located within easy reach and clearly visible to an operator standing on the ground. The control panel(s) shall be easily removable or easily acessible from the rear for maintenance purposes. The control panel(s) shall contain a pumping selector control valve for underwing and overwing fueling; a hose reel shutoff valve; underwing nozzle pressure gage; filter/separator differential pressure gage; fuel quality monitor differential pressure gage; pump discharge pressure gage; control panel light; and other controls considered necessary for underwing and overwing operation. On or near this control panel, provision shall be for a pumping selector control valve for defuel position; bottom loading controls; and other controls considered necessary for overwing, defueling, and bottom loading operations. All controls shall be arranged for maximum convenience to the operator.

3.12.16 Pressure gages. All pressure gages shall incorporate restriction plugs as required to dampen oscillations and provide stability of the indicating needle.

* 3.12.17 Defuel system. The truck shall be equipped with a defuel system. The defuel system shall utilize a suction inlet stub/defuel manifold, and a special hose kit and attachment as described in paragraphs 3.12.17.1 and 3.12.17.2 for receiving fuel from the aircraft. The defuel system shall discharge into the product fuel tank through the internal valve by way of the bottom loading stub described in paragraph 3.14. Fuel from the aircraft shall be brought on board the truck by the truck's pump through the suction stub and circulated through the filter/separator, the relaxation tank, meter, hose reel and hose, and into the product fuel tank throught the underwing nozzle

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connected to the turck's bottom loading system. The defuel system design shall incorporate controls that will keep the pump from cavitating when defueling aircraft at slow defuel rates. A manual wheel operated gate regulating defuel valve is acceptable as a means to control flow and preclude pump cavitation. Also, pumping selector control valves for the defuel system operation shall be located in the area of the defuel system stub and the bottom loading stub. The defuel system regulated flow rates shall be as specified in 3.12.3(d). There shall be no common section of piping that will, after defueling operation, contain unfiltered fuel that would require purging before delivery of product fuel through the underwing refueling system.

* 3.12.17.1 Suction stub/defuel manifold. A suction stub shall be incorporated into the refueler piping design that will allow for its use in the defueling operations. The suction stub shall be located near the bottom loading manifold in the front pumping compartment at the driver's side of the truck and shall be provided with 3 inch flange as specified in MS 27031-3 for attachment of the truck half of the emergency breakaway coupler. The piping for the suction stub shall terminate into the tank-to-pump suction line before the strainer.

* 3.12.17.2 Defuel hose kit and attachments. A 50 foot length of 2 inch diameter hose with an AERO QUIP UNISEX nozzle adapter on one end and with an automatic, gimbal, actuated emergency breakaway coupler on the other end, shall be provided for the defuel system. The hose shall be in accordance with MIl-H-27516D with internal bonding/tension wire for electrical continuity. The hose shall be banded to the nozzle end adapter and to the emergency breakaway coupler with stainless steel bands. The truck half of the emergency breakway coupler shall match up with 8 mounting bolts at the 3 inch suction stub flange described in paragraph 3.12.17.1. The nozzle end of the hose shall be able to be connected with various adapters that are incorporated on a RDF nozzle, a D-1 nozzle, and an overwing nozzle that will contain a 30 inch long length spout assembly for defueling auxiliary fuel tanks (tip tanks). A complete defuel hose assembly kit with 50 foot length of 2 inch hose, UNISEX nozzle adapter and emergency breakaway coupler shall be provided with the truck and shall be equal to AERO QUIP part number AE-706994-2. A separate RDF nozzle equal to AE-83277R (without sampler) shall also be provided and an overwing nozzle with long length spout assembly equal to AERO QUIP part number AE-83958R shall be provided as accessory items. Provision shall be made to stow the complete defuel hose kit assembly and other hose end adapters and nozzles in an open storage compartment located under the product fuel tank, aft of the left rear fender of the truck. The floor of the storage compartment shall be designed so that spilled fuel will not accumulate in the compartment. The accessory items shall be protected against road dirt.

* 3.12.18 Piping. The pumping system piping layout shall be designed to facilitate maintenance on the pumping system and removal of components with minimum disassembly of piping. Piping sections shall be easily removable for maintenance and repair. The maximum practicable extent, piping connections to major and minor components shall be of the groove type. The piping shall be designed to facilitate drainage. The large pipe sections, and pipe sections which could be subjected to excessive movement during operation, shall be

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firmly mounted or braced. Piping wall thickness shall be a minimum of schedule 40 in both the pumping system and the bottom loading system. Piping shall be constructed of ASTM B209 aluminum alloy No. 6061, temper T4 or T6, or equal. Tank inlet and discharge piping shall comply with CFR 49, paragraph 178.340-8 of MC306 for accident damage protection against escape of contents. All piping shall be protected at all locations where it passes through frames and bulkheads.

3.12.19 Tubing and conduit. Wherever possible, static pressure and gage-sensing tube lines shall be installed in groups and mounted on blocks which provide for separation of the lines and mounting of multiple lines in one place on any flat surface.

3.13 Grounding cable and reel. A static grounding cable and reel shall be provided. The grounding cable shall consist of at least No. 12 American Standard Wire Gage Stranded Wire, clear plastic covered. The cable shall be 75 feet long and shall terminate in a wye. Each leg of the wye shall be 15 feet long. Total wire length is 90 feet. One end of the wye cable shall be provided with a spring type, heavy-duty "alligator" clamp and the other end of the wye cable shall be provided with both a plug insert for plug and jack type bonding and an alligator clamp. The cable shall be contained in a spring loaded, interrupted gear, ratchet type reel. A cable outlet guide shall be provided to prevent the grounding cables from tangling or kinking when being extended or retracted in any direction not exceeding 90° from axis of cable outlet guide. A cable stop, which will prevent damage to the clamps due to rapid rewind, shall be provided.

* 3.14 Bottom loading system. The truck shall be equipped with a bottom loading system that shall allow the tank to be filled at any flow rate up to 600 gpm. The bottom loading system shall be in accordance with API RP1004 bottom loading for MC306 tank vehicles, except as specified herein. A pneumatic type interlock (with air brake system) shall be provided to prevent movement of the truck when the bottom loading adapter is engaged for loading.

* 3.14.1 Inlet manifold. The bottom loading inlet manifold shall extend from the inlet receptacle to the bottom of the tank through the emergency internal issue valve as described in paragraph 3.12.4.2. This valve shall be located in an area separated by baffles from the inlet to the tank-to-pump suction line. Fuel input when bottom loaded shall be effectively controlled by means of a deflector to minimize turbulence or spray of product. A manual emergency valve control lever shall be provided accessible to the operator for positive ON and OFF control of the system.

* 3.14.1.1 Inlet receptacle. The inlet receptacle shall be located on the left or driver's side of the truck in a convenient location to allow attachment of the pressure servicing nozzle and hose. The receptacle shall mate with a type D-1 underwing/pressure servicing type nozzle (see 3.12.8.2.1), and incorporate a high-strength adapter, conforming to MIL-A-25896, and MS24484, part No. MS24484-2. The flange adapter shall be fitted with a product coding or selection ring similar or equal to model GTP-540A, Gammon Technical Products, Inc., Manasquan, NJ, which shall prevent

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connection between adapter and nozzle equipment having different fuel assignments. The adapter shall be designed for four positions of product selection. A dust cover shall be furnished. The inlet receptacle shall be designed to open by attachment and operation of a type D-1 nozzle. The receptacle shall be substantially mounted to avoid failure due to the heavy nozzle and hose.

* 3.14.2 Loading an unloading vent. For normal venting under loading and unloading conditions with dome cover closed (bottom loading, defuel operation, and fuel dispensing), an automatic pressure-vacuum-sensing vent equal to AERO QUIP AE-831480, shall be mounted in the top of the tank. The vent operation shall limit the vacuum of 1 psi and tank pressure to 3 psig based on maximum product fuel transfer rates. Opening and closing of the vent shall be controlled by pressure when loading and vacuum when unloading the product fuel. The vent shall be designed to prevent loss of liquid through the vent in case of vehicle upset.

* 3.14.3 Liquid level control. An intrinsic, safe, electronic type warning system shall be provided in the truck for liquid level control and overfill prevention when filling the vehicle's product fuel tank. The complete warning system shall contain the necesary on board components that shall provide a positive, self-checking, continuously monitored visual and audible tank fill warning system and through an external truck mounted socket, provide an electronic signal for fill stand on and off control of fuel loading. These truck mounted components shall consist of the minimum following devices as described in paragraph 3.14.3.1.

* 3.14.3.1 Liquid level system components. For liquid level control and over fill protection when loading fuel into Navy tank trucks, an electronic type probe mounted in the top of tank near the manhole shall be provided. This electronic probe shall detect the preset liquid level in the tank by either a fiber optic or thermistor type sensor probe. The probe shall be connected to a power control monitor for audio and visual warning, and shall provide a positive indication when probe is wet or dry. Audio shall be from intermittent to continuous sound with a means to select two ranges of 60 decibel (dB) and 120 dB. Visual shall be from green to red light. Installation of the control monitor shall be in the area adjacent to the bottom loading stub located within easy reach and clearly visible to the operator standing on the ground. The probe shall also be connected through a wire protected metal conduit to an external socket mounted on the truck near the bottom loading controls to connect the truck (by mating plug) to the fill stand for control of fuel loading. The socket shall be of the four prong type, (ATA-4) with dust cap, compatible with Navy fill stand loading systems ("SCULTROL" or equal). The socket connection shall be protected against inadvertent truck drive away from the fill stand. In the event of a tank overfill situation or the desire to control fuel loading of the truck at the fill stand, the system shall be able to be shut down by the fill stand controls or the truck operator shall be warned for manual shut down when the product fuel touches the electronic probe in the truck's tank. The probe length shall be ajustable between 93 percent and 95 percent of tank capacity to establish fuel contact and to initiate preset shut-down of product fuel

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loading or to sound the alarm. An on-board control unit shall be provided to operate the electronic probe and the signal conversion circuits for audio and visual warning. Accessory power shall be provided to the control unit by the trucks' battery system by a separate control switch (not connected through the trucks' engine ignition switch).

* 3.14.4 Circulation of product fuel. A means shall be provided in the design of the bottom loading system to permit product fuel circulation from the filled tank through the filter/separator and fuel quality monitor and back into the tank through the internal valve. Product circulation is to be performed at standard rpm speed settings where flow rates can be measured and differential pressure readings are meaningful.

* 3.15 Night servicing light. The truck shall be provided with an automotive type, 6-inch sealed beam, roof or side mounted type spotlight. The spotlight shall be located near the front corner (driver's side) of the cab roof and shall swivel to allow the light to illuminate the ground area to the rear and adjacent to the left side fueling compartments. Controls shall extend inside the vehicle cab for ON-OFF control and movement of the lamp housing through horizontal and vertical positions.

3.16 Fire extinguishers. Two 30-lb dry chemical extinguishers shall be furnished. These extinguishers shall conform to O-E-915, size 30, type II, class 2, and shall be approved for use with Purple-K-Powder (P-K-P). The extinguishers shall be charged with potassium dry chemcial P-K-P agent conforming to O-D-1407. Each extinguisher shall be mounted with a suitable truck mounting carrying bracket, easily and quickly operated. One fire extinguisher shall be mounted vertically on each side of the truck. Protection shall be provided from splashing of mud and water. The fire extinguishers shall be readily accessible to personnel standing on the ground.

3.17 Operational marking.

3.17.1 Instruction plates. The truck shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.17.2 Diagrams. Schematic diagrams of the piping, including the bottom loading system and electrical system, shall be provided. Each valve, switch, and other pertinent component on the diagram shall be properly identified to correspond to the markings on like parts of the truck. Schematic diagrams shall be provided on metal plates or plastic encapsulated plates and shall be of a material which will be durable and remain legible for the life of the equipment. Diagrams shall be located and attached as specified in 3.17.1.

3.17.3 Operating instructions. Operating instructions shall be clearly and concisely stated referencing identification indicated on the wiring and piping diagrams. Operations which affect safety of the pumping system shall

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be listed under the title CAUTION at the top of the plate. The word CAUTION shall be larger than the operational instructions. Specific instructions shall be listed separately for each pumping system operating mode, with the pumping system mode as the title for the various sections on the plate.

3.17.4 Bottom loading plate. A concise, step-by-step operating instruction plate for the bottom loading system shall be permanently mounted on or adjacent to the bottom loading control panel. Operations which affect safety shall be listed under the title CAUTION at the top of the plate. The word CAUTION shall be larger than the operational instructions.

3.17.5 Identification plates. Valves, swithces, levers, and controls used in the operation of the truck shall be properly identified by nonferrous identification plates. The data plates shall incorporate a numerical or alphabetical system of identification and an indication of the relative position of the lever or control handle, such as ON and OFF.

3.17.6 Filter/separator element and fuel equality monitor change criteria data plates. An element change criteria data plate shall be mounted adjacent to the filter/separator pressure gage specified in 3.12.5.5. The data plate shall be entitled FILTER/SEPARATOR ELEMENT CHANGE CRITERIA. The plate shall indicate that the elements shall be changed at 20 psi pressure differential, every 3 years, or when the fuel quality monitor triggers (shuts off the flow), whichever occurs first. A second data plate, similar to above, shall be mounted adjacent to a differential pressure gage to be provided and used for the fuel quality monitor. The data plate shall be entitled FUEL QUALITY MONITOR CHANGE CRITERIA. The plate shall indicate that the elements shall be changed at 20 psi pressure differential, or when the unit triggers (shuts off the flow), whichever occurs first.

3.18 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.19 Identification marking and nameplates. Identification marking and nameplates shall be in acordance with MIL-STD-1223, as required for the Navy. Transportation data plates shall be furnished and shall be inscribed with all pertinent transportation date, including a silhouette of the truck showing the center of gravity. A standard Navy nameplate, to be furnished by the contracting officer, shall be inscribed by the contractor with the pertinent data in the blank spaces provided: TRUCK, TANK, AIRCRAFT REFUELING, 5,000 GALLON. The contractor shall securely affix the plate inside the truck cab, with nonferrouss screws, bolts, or rivets of not less than 1/8-inch diameter.

3.20 Cleaning, treatment, and painting. The exterior surfaces of the refueler truck body shall be cleaned and treated in accordance with MIL-T-704, type A. The exterior surfaces to be painted shall be primed with a zinc

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chromate epoxy applied to a dry file thickness of 1.5 - 2.5 mils. The primer used shall be compatible with the final paint system. The primer paint shall be polyurethane enamel or equal, applied to a dry film thickness of 1.8 - 2.2 mils. The final finish shall be applied to the exterior surfaces of the body tank, and interior and exterior of defuel housing and pumping compartment. The finish polyurethane enamel shall be equal to DUPONT "IMRON," DITZLKER "DURETHANE," or SHERWIN-WILLIAMS "POLASOL," brand products. Treatment and painting of the truck chassis and cab shall be in accordance with MIL-STD-1223, except the exterior color of the entire truck (body and chassis cab) shall be gloss yellow matching color chip No. 13538 of FED-STD-595. Both front and rear bumper plates shall be painted black No. 17038, then 4-inch stripes of yellow reflectorized sheeting conforming to L-S-300 shall be applied 4 inches apart at the incline of 45° from the left and right of vertical to simulate an inverted V-pattern of yellow and black stripes.

3.21 Rustproofing. When specified (see 6.2), the truck cab, chassis, and chassis underside shall be rustproofed in accordance with FED-STD-297.

* 3.22 Servicing and adjusting. Prior to acceptance of the truck by the Government inspector, contractor shall service and adjust each truck and its mounted equipment for operational use, including at least the following: Focusing of lights; adjustment of engine; adjustment of electrical and brake system; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear, and mounted equipment with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

3.23 Workmanship.

3.23.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.23.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tightended to prescribed torque.

3.23.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

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3.23.4 Welding. All welders employed in the fabrication of the product fuel tank, and the subassembly and assembly of the product fuel tank to the truck chassis, shall be certified in accordance with MIL-STD-248. All welding procedures shall be in accordance with the highest standards of industry. Adequate preparation of the metal prior to welding of tank shell, tank end plates, baffles, restraining devices and support brackets, appurtenances, components, and accessories shall be provided. Surface of parts welded shall be free of dust, dirt, scale, paint, grease, and other foreign matter. Aluminum alloy electrodes or filler metal used in the welding process shall be high quality metal of the correct size and alloy composition to maintain the proper strength, ductility, and resistance to weld cracking by keeping base metal alloy dilution, at the weld deposit, to a minimum. Care must be taken during storage and use of filler metal to prevent contamination that would cause poor welds. Weld penetration shall be such as to provide tranference of maximum design stress through the base metal juncture. Weld beads shall be thoroughly cleaned and spattering shall be removed from the tank.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination. Each truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.2.1 First article inspection. The first article inspection shall be performed on truck when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.2, the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

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4.3 Classification of tests. Testing of the trucks shall be classified as follows:

a. First article tests (see 4.5).
b. Acceptance tests of production units (see 4.6).

4.4 Test conditions.

4.4.1 Apparatus. Apparatus used in conjunction with testing specified herein shall be of laboratory precision type and a classification record shall be maintained, and made available to the Government for examination showing that the apparatus has been calibrated within 3 months preceding date of tests. All pressure readings in the first article shall be taken by using transducers and recording type oscillograph. Records shall be made available to the Government for examination showing that the transducers have been calibrated.

4.4.2 Liquid flow rates. Liquid flow rates shall be measured to determine conformance to 3.12.7.

4.4.3 Test fluid. The test fluid used for all hydrostatic tests and for all liquid flow tests shall be Stoddard's solvent, commercial jet fuel, or JP-5.

4.4.4 Data. During testing of the pumping system, at least the following data shall be recorded:

a. Pressure at pump inlet - inches Hg.
b. Pump discharge pressure - psig.
c. Pressure loss through meter - psig.
d. Pressure loss through filter/separator - psig.
e. Pressure loss through hose - psig.
f. Pressure loss through fuel monitor - psig.
g. Slug control operations.
h. Fuel flow rates - gpm.
i. Engine speed - rpm.
j. Pump speed - rpm.

4.4.5 Cleaning. Prior to the beginning of any tests, all trucks, including the first article, shall be subjected to the following cleaning procedure: Each truck shall be carefully inspected for evidence of contaminants in the tank and fuel system, such as loose, spattered or excess solder, metal chips, or other debris. Test fluid, as specified in 4.4.3, shall be pumped through the system for a minimum of 10 minutes at least 100 gpm. At the end of the 10 minutes, strainers and low point drains shall be examined for evidence of contaminants. If any trace of contaminants is found, the recirculation of test fluid at 100 gpm will continue for 10 minute periods until all trace of contaminants have been removed. Strainers and low point drains will be examined at the end of each 10-minute period. When all trace of contaminants have been removed, new and unused elements shall be installed in the filter/separator and fuel quality monitor, and utilized in

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all performance tests specified herein. It is intended that only one set each of filter/separator elements and fuel quality monitor elements be used in testing the first article and all production units.

4.5 First article tests. When first article is required, the first article shall be tested as specified in 4.5.1 through 4.5.15. First article tests shall be performed by the coontractor under the direction and in the presence of Government representatives. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all trucks until corrective action has been taken.

4.5.1 Tank air test. To determine conformance to 3.10.1, the truck cargo tank shall be subjected to a static air pressure test at 3 psig for a period of not less than 30 minutes. A soap solution shall be applied on the exterior of the tank at all welds and joints to facilitate the detection of any leaks.

4.5.2 Piping system hydrostatic tests. To determine conformance to 3.12.1, all components of the piping system, including the internal valve, all pipe fitting, hose nozzles; and other components through which fuel passes, shall be hydrostatically tested, with the test fluid as specified in 4.4.3, at 150 percent of design pressure for a period of at least 30 minutes.

4.5.3 Vent operations test. The venting equipment shall be tested for conformance to the requirements of 3.10.6 and 3.14.2. The vents shall relieve positive and negative pressures and prevent steady stream leakage.

4.5.4 Flow test. To determine conformance to 3.12.3, test fluid as specified in 4.4.3, at normal ambient temperature, shall be pumped from the tank to a point which is 20 feet above the pump base at maximum flow rate for a minimum period of 10 minutes. Flow shall be through the underwing servicing nozzle. Nozzle pressure shall not exceed 50 psig $\pm$5 psig. Following this, a second flow test shall be performed using 1-1/2 inch hose and overwing nozzle. Flow rates shall be of at least 100 gpm for a minimum of 10 minutes. In all flow tests, a suitable shutoff valve may be installed at the outlet of the nozzle to simulate the 20-foot head, if desired. Maximum vacuum at the pump suction shall be 7 inches Hg. During flow tests, recording of data shall be in accordance with the test data required in 4.4.4.

4.5.5 Bottom loading system test. The truck cargo tank shall be filled to its rated capacity through the bottom loading connection to determine conformance to the requirements of 3.14. While the cargo tank is being filled, the liquid level control and circulation of product fuel system shall be operable and mounitored to determine conformance to 3.14.3, 3.14.3.1, and 3.14.4.

4.5.6 Valve closure test. While pumping at maximum gpm through the pressure servicing nozzle and with the nozzle raised at a minimum of 10 feet above ground level, at least 10 valve closures at a 0.5 second closure time shall be made to determine conformance to 3.12.3c. The total pressure buildup at the nozzle, including the 50 psig, the 10-foot head, and any surge pressure, shall not exceed 120 psig.

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4.5.7 Pump suction. To determine conformance to 3.12.2, the pump shall be tested to determine capability of producing a minimum of 18 inches Hg when loading the truck from an outside source or defueling aircraft.

4.5.8 Hose reel test. To determine conformance to 3.12.8, fluid handling sections of hose reel shall be given hydrostatic tests of 150 percent of design pressure to determine strength and porosity. Test fluid shall be as specified in 4.4.3. Pressure shall be applied in 50 psig increments. Pressure will be held for 10 minutes at each increment. There shall be no leaks. The hose reel shall be subjected to a vacuum test of at least 15 inches Hg for a minimum of 10 minutes. There shall be no leaks.

4.5.9 Emergency dry break connector. An operational test of the emergency dry break connector system required on the hose reel as described in paragraph 3.12.8.1 shall be conducted to verify performance requirements.

4.5.10 Defuel test. The 50 foot 2 inch non-collapsible defuel hose kit shall be attached to a storage tank or the truck cargo tank, to determine conformance to 3.12.17. The selector valve shall be set for the defuel mode. A flow rate of not less than 80 gpm nor more than 100 gpm shall be maintained for at least 5 minutes. The level of test fluid in the storage tank or truck cargo tank shall be no higher than 6 feet above the truck pump at the beginning of the test.

4.5.11 Tests for electromagnetic interference suppression. The truck shall be tested to determine conformance of 3.3.5. In lieu of tests to determine conformance to SAE J551, the manufacturer may prepare a certification that the truck meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.5.12 Weight check. The weight of each wheel shall be measured with the truck standing on a flat surface, first empty and then with cargo tank loaded with fuel equivalent to the rated load on 5,000 gallons at 6.6 lb per gallon, and with the pumping system fully flooded as during normal operation. The load on each side of the bogie assembly will be acceptable in lieu of the individual wheels of the bogie.

4.5.13 Road tests. The truck shall be driven 200 miles, consisting of two cycles of 100 miles. The first cycle shall be run with the cargo tank half-filled with fuel. The second cycle shall be run with the cargo tank filled to capacity. Each cycle shall consist of the following phases:

a. A total of 75 miles at a speed of 25-35 mph over highways or improved roads.
b. A total of 25 miles at a speed of 10-25 mph over gravel or secondary/unimproved roads.

During the complete road test, at least 10 short radius right turns, 10 short radius left turns, and 10 sudden stops shall be conducted. Throughout the test, the truck shall be observed as to its traceability, ease of handling,

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tendency to yaw, or sway. Upon completion of the road tests, the truck shall be inspected for misalingment, distortion, or other damage sustained as a result of the testing. Undesirable, roadability feature and any damage resulting from tests shall be cause for rejection.

4.5.14 High speed. The truck shall be loaded to rated capacity. It shall be demonstrated that the truck can achieve a speed of 45 mph on level paved highway within 1.5 miles from a dead stop.

4.5.15 Service brake test. The truck shall be loaded to rated capacity. The truck shall be subjected to three emergency stops in rapid succession from 20 mph on clean, dry, smooth, level, paved surface free from loose material. Each stop shall be accomplished in not more than 1-minute intervals. The truck shall stop within 35 feet from the point the brakes are applied. A time-distance recorder shall be used and shall positively identify the point where the brakes are initially applied. This test shall be conducted not less than four times over duration of the road test.

4.5.16 Test records. Contractor's records of all first article tests and inspections, giving the results of said tests and inspections to determine compliance with the requirements of this specification, shall be kept complete and available to the Government's representative and the contracting officer. Test and inspection records shall be signed and approved by a person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory, satisfactory to the Government, capable of conduction required tests to determine compliance with all the requirements of this specification.

4.5.17 Production sample. Upon acceptance of the first article, it shall remain at the manufacturing facility as a production sample, and be the last truck shipped on the contract. The first article shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The elements in the filter/separator and fuel quality monitor utilized in the first article tests shall be removed and replaced with new and unused elements. The contractor shall maintain the first article in a serviceable condition for the duration of the the contract.

4.6 Acceptance test of production units. Each production unit shall be subjected to the tests specified in 4.4.5, 4.5.1, 4.5.2, and 4.5.5. In addition, a modified flow test shall be performed to determine that all parts are functioning properly. A minimum of 3,000 gallons of test fluid as specified in 4.4.3 shall circulated through the system at maximum flow rate. At the end of the flow test, all test fluid shall be completely drained from the tank, filter/separator, and all fluid handling piping and components. Upon acceptance of each production unit, the elements in the filter/separator and fuel quality monitor utilized in the production unit tests shall be removed and replaced with new and unused elements.

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4.7 Packaging inspection. The preservation of the trucks shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Unless otherwise specified (see 6.2), the complete truck shall be preserved and packaged in accordance with the supplier's standard practice. When specified (see 6.2), the complete truck shall be preserved in accordance with the (mobile) requirements of MIL-STD-281, with the level of preservation and packaging as specified.

6. NOTES

6.1 Intended use. The truck covered by this specification is intended for use at established Naval Air Activity for fueling aircraft both in and outside the continental limit of the United States.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
c. When a particular measurement system is required (see 3.6.1).
d. When lifting and tying down attachments are required (see 3.3.8).
e. When air transportability is required (see 3.3.9).
f. When an inflated tire is mounted on a spare wheel (see 3.9.11.4).
g. When tools are required (see 3.9.21).
h. When vapor recovery system is required (see 3.10.5.1).
i. When rustproofing is required (see 3.21).
j. When preservation in accordance with MIL-STD-281 is required, and the level of preservation-packaging required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Performance prediction. Complete copies of SAE Truck Ability Prediction Procedure form and computation for maximum geared speed requirement for the vehicle model furnished under contract should be submitted as

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specified in the contract. Unless other conditions are cited in this contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry, air. The factors to be used in prediction truck ability (see 3.8.1.1) are established as follows for the corresponding SAE J688, Truck Ability Prediction Procedure Tables:

Table 1	-	Tire factor. This factor must relate to size of tires furnished by the contractor in accordance with this specification.
Table 2	-	Altitude factor. 1.00
Table 3	-	Rolling factor. 1.003
Table 4	-	Area factor. 0.197
Table 5	-	Velocity factor. 85.7
Table 6	-	Altitude factor. 1.00
Table 7	-	Chassis friction hp. Use applicable power unit gvw and the rpm (to nearest 100 revolutions) which is required for a geared speed of 35 mph. For engine speed beyond the range of this table, the following formula shall be used: FHP = 1 + (0.0000002 rpm) (gvw).
Table 8	-	Grade factor. 1.07
Table 8A	-	Correction factor. Not required.
Table 9	-	Road factor. 0.0

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Navy - YD

Preparing Activity:

Navy - YD

(Project 2320-NC48)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-28612E	TRUCK, TANK: ARICRAFT REFUELING, 5000 GALLON, AVGAS OR JET FUEL 6 BY 4, DED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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